

1. An AI speaker is a typical use case for speech processing.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

Answer Explanations:

Nothing

2. Huawei's all-scenario AI solutions can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

Answer Explanations:

Nothing

3. Typically, in a network that uses sigmoid, the gradient decreases to 0 within five layers, making the network difficult to train.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

Answer Explanations:

Nothing

4. When you train a deep neural network with the softsign function, if the network is very deep, an increasing number of back-propagated gradients fall into the saturation area, making the gradient module smaller and finally close to 0, and the weight cannot be updated.

☐ True

☒ False



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 Sorry, your answer was wrong. My Score: 0 Point(s) Answer: True

[Feedback on Answer](#)

Answer Explanations:

Nothing

5. The tanh function can effectively solve the vanishing gradient problem.

☐ True

☒ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: False

Answer Explanations:

Nothing

6. L1 regularization is referred to as weight decay.

☒ True

☐ False

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: False

Answer Explanations:

Nothing

7. If the hyperparameter search space is large, random search is better than grid search.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

Answer Explanations:

Nothing

8. In a random search, each setting item is sampled from possible parameter values to find the optimal parameter subset.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

Answer Explanations:

Nothing

9. Assuming a dataset contains the areas and prices of 21,613 housing units in a city, you can use a classification model to predict the prices of other housing units in the city.

☐ True

☒ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: False

Answer Explanations:

Nothing

10. Batch normalization is a technique used in deep learning to optimize training. It reduces the dependence on initial weights and learning rates, resulting in improved model performance.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

[Feedback on Answer](#)

Answer Explanations:

Nothing

11. In regression tasks, the most commonly used loss function is cross entropy cost.

☐ True

☒ False

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✓ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: False

Answer Explanations:

Nothing

12. In data parallelism, the data of each training batch is split into N parts, forward propagation is performed across computing units, and the loss value calculated by each card is aggregated on the master node.

☒ True

☐ False

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: False

[Feedback on Answer](#)

Answer Explanations:

Nothing

13. In pipeline parallelism, the parameters of each layer of a model are split into multiple parts and distributed to multiple computing units. After the forward computation is complete, the gradients are combined on the master computing unit in the original splitting mode.

☒ True



☐ False

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 Sorry, your answer was wrong. My Score: 0 Point(s) Answer: False

[Feedback on Answer](#)

Answer Explanations:

Nothing

14. Model fine-tuning is to adjust the model structure and the number of trainable parameters based on the pre-trained model, and to use more diverse data for training.

☒ True

☐ False

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✘ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: False

Answer Explanations:

Nothing

15. The core idea of adapter tuning is to add adapter modules to specific layers of a pre-trained model. These modules usually consist of two small neural networks: one for dimensionality reduction and the other for augmentation.

☒ True

☐ False

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: True

[Feedback on](#)

Answer Explanations:

Nothing

16. Without considering any regularization terms, the support vectors of a support vector machine (SVM) are composed of ().

- ☐ A. Points on the separating hyperplane
- ☐ B. The points farthest from the separating hyperplane
- ☒ C. The points closest to the separating hyperplane
- ☐ D. Points of a certain type

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: C

Answer Explanations:

Nothing

17. During neural network training, which of the following values is continuously updated using the gradient descent method to minimize the loss function?

- ☐ A. Hyperparameters
- ☐ B. Feature value
- ☐ C. Number of samples
- ☒ D. Parameters

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: D

Answer Explanations:

Nothing

18. The ReLU function is commonly used in deep learning neural networks. Which of the following is the value range of this function?

☐ A. $[0, +\infty)$

☒ B. $[0, 1]$

☐ C. $[-1, 1]$

☐ D. $[-1, 0]$

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: A

Answer Explanations:

Nothing

19. The sigmoid activation function is monotonic and continuous, has bounded outputs, and makes the network easy to converge. It was popular for a while, but when the network is deep, which one of these problems is sigmoid associated with?

- ☐ A. Gradient reduction
- ☒ B. Vanishing gradient
- ☐ C. XOR Problem
- ☐ D. Overfitting

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: B

[Feedback on Answer](#)

Answer Explanations:

Nothing

20. Which of the following functions can be used to alleviate the vanishing gradient problem?

- ☐ A. Sigmoid
- ☐ B. Tanh
- ☐ C. Softsign
- ☒ D. ReLU

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✓ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: D

Answer Explanations:

Nothing

21. If a 32 x 32 image is input and a 5 x 5 kernel is used for convolution with the stride of 1, the size of the output image is ().

☒ A. 28 x 28

☐ B. 29 x 29

☐ C. 28 x 23

☐ D. 23 x 23

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✓ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: A

Answer Explanations:

Nothing

22. A manufacturer wants to produce virtual assistants for hospital use. For virtual assistants, which of the following technologies is used for voiceprint recognition?

- ☒ A. Speech recognition and processing
- ☐ B. Image recognition and processing
- ☐ C. Expert system and knowledge graph
- ☐ D. Image generation and enhancement

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: A

Answer Explanations:

Nothing

23. Which of the following is NOT a use case for natural language processing?

- ☐ A. Public opinion analysis
- ☐ B. Machine translation
- ☐ C. Text classification
- ☒ D. Image recognition

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: D

Answer Explanations:

Nothing

24. Which of the following is NOT a recurrent neural network?

☐ A. RNN

☐ B. LSTM

☒ C. CNN

☐ D. RNN

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✓ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: C

Answer Explanations:

Nothing

25. Within the broad landscape of large language models built on the Transformer architecture, which roadmap does GPT belong to?

- ☐ A. Encoder-Only
- ☒ B. Decoder-Only
- ☐ C. Encoder-Decoder
- ☐ D. Decoder-Encoder

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✔ Congratulations! Your answer was correct.

My Score: 16 Point(s)

Answer: B

Answer Explanations:

Nothing

26. Which of the following statements about different optimizers is false?

- ☐ A. RMSprop solves the problem of the Adagrad optimizer ending too early.
- ☐ B. Compared with RMSprop, Adagrad is more sensitive to gradient changes.
- ☐ C. Both Adagrad and RMSprop can set adaptive learning rate for parameters.
- ☒ D. Adam requires bias correction for initial iterations.

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Sorry, your answer was wrong.

My Score: 0 Point(s)

Answer: B

Answer Explanations:

Nothing

27. A computer uses labeled images to learn and distinguish between images of apples and those of pears. What is the name of this learning method?

- ☒ A. Supervised learning
- ☐ B. Unsupervised learning
- ☐ C. Semi-supervised learning
- ☐ D. Reinforcement learning

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: A

Answer Explanations:

Nothing

28. Which of the following statements is true about classification models and regression models in machine learning?

- ☐ A. For regression problems, the output variables are discrete values. For classification problems, the output variables are continuous.
- ☐ B. The most commonly used indicators for evaluating regression and classification problems are the accuracy and the recall rate.
- ☒ C. Overfitting may occur in both regression and classification problems.
- ☐ D. Logistic regression is a typical regression model.

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Congratulations! Your answer was correct.

My Score: 16 Point(s)

Answer: C

Answer Explanations:

Nothing

29. Which of the following statements about support vector machines (SVMs) is false?

- ☐ A. SVMs are classification models. Their basic model is the linear classifier that maximizes the width of the gap between the two categories in the feature space.
- ☐ B. SVMs also have a kernel trick, which makes them non-linear classifiers.
- ☒ C. In the case of linear inseparability, non-linear mapping algorithms are used to convert the linearly inseparable samples of low-dimensional input space into samples of high-dimensional feature space. In this way, samples become linearly separable. ✖
- ☐ D. SVMs only apply to linear classification.

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: D

[Feedback on Answer](#)

Answer Explanations:

Nothing

30. Which of the following is NOT an AI deep learning framework?

- ☐ A. TensorFlow
- ☐ B. Scikit-learn
- ☐ C. Pytorch
- ☒ D. Caffe

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✗ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: B

Answer Explanations:

Nothing

31. Which of the following are used in TensorFlow to describe the computation process?

☐ A. Parameter

☐ B. Session

☒ C. Data flow

☐ D. Tensor

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: C

Answer Explanations:

Nothing

32. If a 32×32 feature map is processed through a pooling layer with a stride of 2, the size of the output map is ().

☐ A. 30×30

☒ B. 16×16

☐ C. 29×29

☐ D. 23×23

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: B

Answer Explanations:

Nothing

33. Which of the following is NOT a characteristic of PyTorch?

- ☐ A. Python first
- ☐ B. Easy to debug
- ☐ C. Extensive ecosystem
- ☒ D. Default use of static maps

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: D

Answer Explanations:

Nothing

34. Which of the following statements is true about the training of large models?

- ☐ A. Foundation models cannot be directly deployed.
- ☒ B. The pre-training of large models requires a large amount of high-quality data.
- ☐ C. Unlabeled data used in the pre-training phase of large models must be used in the fine-tuning phase.
- ☐ D. Large models have emergent abilities, and therefore they do not need to be aligned with human values.

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s) Answer: B

Answer Explanations:

Nothing

35. Tensor parallelism is a common method for parallel training of large models. Which of the following types of model data is split to different devices?

- ☐ A. Gradient
- ☐ B. Optimizer
- ☒ C. Input data
- ☐ D. Model parameters

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: D

Answer Explanations:

Nothing

36. What are the layers used in Transformer to capture time dependencies within a sequence?

- ☐ A. Residual layer
- ☒ B. Self-attention layer
- ☐ C. Location code
- ☐ D. Feed-forward network layer

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37. During the calculation of the attention mechanism, which of the following are required for similarity calculation to obtain the attention score?

☒ A. Query

☐ B. Weight

☒ C. Value

☒ D. Key

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: AD

Answer Explanations:

Nothing

38. Which of the following are common evaluation metrics for classification tasks?

☒ A. Accuracy

☒ B. Precision

☒ C. Recall rate

☐ D. Mean squared error

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✔ Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: ABC

Answer Explanations:

Nothing

39. Which of the following hyperparameters may be defined during model training?

- ☒ A. Batch size
- ☒ B. Learning rate
- ☒ C. Optimizer type
- ☐ D. Loss value

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✔ Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ABC

Answer Explanations:

Nothing

40. Optimizers are often used to accelerate training of deep learning neural networks. Which of the following are deep learning optimizers?

☒ A. Momentum

☒ B. Adagrad

☒ C. RMSprop

☒ D. Adam

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 Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ABCD

Answer Explanations:

Nothing

41. Which of the following statements about some of the subfields of AI are true?

- ☒ A. Computer vision is a science that studies how to make computers "see" things. ✓
- ☒ B. Speech processing is a general term for different speech processing methods and techniques, including the study of vocalization, the collection of statistics related to speech signals, speech recognition, speech synthesis, and speech perception. ✓
- ☒ C. Natural language processing studies how to use computer technology to understand and use human language. ✓
- ☐ D. Autonomous driving does not require speech processing or computer vision.

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✓ Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ABC

[Feedback on Answer](#)

Answer Explanations:

Nothing

42. Which of the following are common activation functions of neural networks?

☐ A. Dropout

☒ B. Sigmoid

☒ C. tanh

☒ D. Leaky ReLU

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✔ Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: BCD

Answer Explanations:

Nothing

43. Which of the following comprise the Adam optimizer?

☒ A. Momentum

☐ B. Adagrad

☒ C. RMSprop

☐ D. Nesterov

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✔ Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: AC

Answer Explanations:

Nothing

44. Which of the following statements about datasets are true?

- ☒ A. A dataset is a collection of data used in machine learning tasks. Each piece of data is called a sample.
- ☒ B. Events or attributes that reflect the particular performance or nature of a sample are called features.
- ☒ C. A training set is a dataset used in the training process, where each sample is referred to as a training sample.
- ☐ D. Learning (or training) is the process of building a model from data.

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: ABCD

Answer Explanations:

Nothing

45. Which of the following statements about data preprocessing are true?

- ☒ A. Data cleansing is a process of filling in missing values, as well as detecting and eliminating noise and exceptions.
- ☒ B. Data dimension reduction aims to simplify data attributes and avoid the curse of dimensionality.
- ☒ C. Data standardization aims to reduce noise data and improve model accuracy by standardizing data.
- ☐ D. Machine learning tasks generate results in the form of model outputs. Therefore, model training is more important than data preprocessing.

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✔ Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ABC

Answer Explanations:

Nothing

46. Which of the following statements about model parameters and hyperparameters are true?

- ☒ A. Models contain both parameters and hyperparameters.
- ☐ B. Hyperparameters are automatically learned by models.
- ☒ C. Hyperparameters are manually set.
- ☒ D. Hyperparameters can be used to control training.

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✔ Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ACD

Answer Explanations:

Nothing

47. Data augmentation can improve model robustness and avoid overfitting. Which of the following are data augmentation methods?

- ☐ A. Batch normalization
- ☒ B. Image contrast change
- ☒ C. Image noise adding
- ☒ D. Random crop

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Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: BCD

Answer Explanations:

Nothing

48. The softmax function is frequently used in classification tasks. Which of the following statements about it are true?

- ☐ A. The softmax function is also called the normalized exponential function.
- ☐ B. The softmax regression model is an algorithm for solving binary classification regression problems.
- ☒ C. The softmax function is the generalization of the binary classification function sigmoid.
- ☒ D. The softmax function is usually used together with the cross-entropy loss function.

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✖ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: ACD

Answer Explanations:

Nothing

49. Which of the following are characteristics of eager execution in TensorFlow 2.x?

☒ A. High performance

☒ B. Deployability

☐ C. Intuitiveness

☒ D. Flexibility

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 Sorry, your answer was wrong. My Score: 0 Point(s) Answer: CD

Answer Explanations:

Nothing

50. The process of developing AI applications includes ().

☒ A. Data preprocessing

☒ B. Modeling

☒ C. Model training

☒ D. Model deployment

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Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: ABCD

Answer Explanations:

Nothing

51. Which of the following statements about large models are true?

- ☒ A. A large amount of data is required for training.
- ☒ B. The model parameter scale is large.
- ☐ C. Training data does not need to be cleansed because this helps models learn how to process error information.
- ☒ D. The generalization capability of large models is stronger than that of small models.

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✔ Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: ABD

Answer Explanations:

Nothing

52. Which of the following fields can large model technologies be applied to?

☒ A. Autonomous driving

☒ B. Robots

☒ C. Translation

☒ D. Finance

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 Congratulations! Your answer was correct. My Score: 18 Point(s) Answer: ABCD

Answer Explanations:

Nothing

53. Which of the following may be included in the training dataset of GPT-3?

☒ A. COCO2017

☒ B. Common Crawl

☒ C. Wikipedia

☒ D. Books1

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 Sorry, your answer was wrong. My Score: 0 Point(s) Answer: BCD

Answer Explanations:

Nothing

54. Which of the following fine-tuning methods do not change the Transformer block structure of a large model?

☐ A. Adapter tuning

☒ B. Prefix tuning

☒ C. Prompt tuning

☐ D. LoRA

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Congratulations! Your answer was correct.

My Score: 18 Point(s)

Answer: BC

Answer Explanations:

Nothing

55. The development of large models requires not only training and fine-tuning, but also evaluation. Which of the following are reasons for evaluating large models?

☐ A. Improving the accuracy of model outputs

☒ B. Evaluating the capabilities of large models

☒ C. Providing reference for enterprises to select models

☒ D. Aligning large models with human values

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✘ Sorry, your answer was wrong. My Score: 0 Point(s) Answer: BC

Answer Explanations:

Nothing

56. The three essential elements for AI development are data, compute, and ____.

1 algorithms

Answer: Algorithm;Algorithms

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s)

Answer Explanations:

Nothing

57. CNN stands for _____.

1 Convolutional Neural Network

Answer: Convolutional neural network

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✔ Congratulations! Your answer was correct. My Score: 16 Point(s)

Answer Explanations:

Nothing

58. A/An ____ is a binary classification model that is based on a linear classifier defined in the feature space. (Enter an abbreviation.)

1

Logistic

Answer: SVM

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✖ Sorry, your answer was wrong. My Score: 0 Point(s)

Answer Explanations:

Nothing

59. The ____ algorithm is based on the idea that if most of a sample's k nearest neighbors in the feature space belong to the same class, the sample also belongs to that class. (Enter an abbreviation.)

1

KNN



Answer: KNN

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 Congratulations! Your answer was correct.

My Score: 16 Point(s)

[Feedback on Answer](#)

Answer Explanations:

Nothing

60. In k-fold cross-validation, the existing data is divided into ____ groups.

1

K

Answer: k

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Congratulations! Your answer was correct.

My Score: 16 Point(s)

Answer Explanations:

Nothing